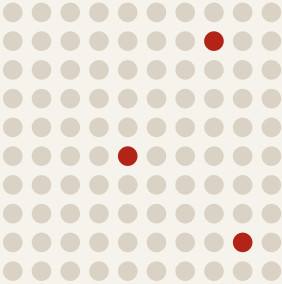
AUGMENTED DECISION · FINAL PROJECT

Flagging bankruptcy *before* it happens

Cost-sensitive screening of Taiwanese firms with LASSO selection and a decision tree



100 FIRMS · 3 GO BANKRUPT

Yassine ERRAJI · William PELTIER · Ben DAESH · Simon AMAR

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THE QUESTION

Should this firm be flagged?

A credit analyst reads a listed firm's financial ratios and must choose one of two actions:

Flag
deny or reprice credit, open a closer review

Do not flag
business as usual

Our goal

A decision rule that is **cheap on average**…

…and that the analyst can **read and question**.

ROADMAP

Four steps

01

Problem formulation

Costs, metric and a first look at the data

02

Variable selection

LASSO keeps 43 of the 94 ratios

03

Model selection

Tune the tree, compare candidates

04

Evaluation

One look at 1,364 unseen firms

PART 01

Problem formulation

What are we predicting, and how do we score a decision?

01

The data

6,819
listed Taiwanese firms

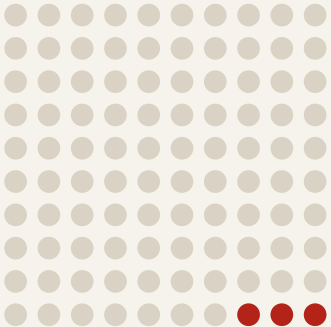
95
financial ratios per firm

11 years
1999–2009, Taiwan Economic Journal

3.2 %
went bankrupt (220 firms)

Source: UCI Machine Learning Repository, *Taiwanese Bankruptcy Prediction*, doi:10.24432/C5004D.

Accuracy is a trap



1 DOT = 68 FIRMS

A model that **never flags anyone** is

96.8 % accurate

…and catches **0 bankruptcies**.

— **We need a metric that knows a missed bankruptcy is worse than a false alarm.**

Mistakes do not cost the same

	Healthy firm	Bankrupt firm
Not flagged	0	10 missed bankruptcy C_{FN}
Flagged	1 false alarm C_{FP}	0

A missed bankruptcy is assumed to cost as much as 10 false alarms.

Stipulated, not estimated: Step 4 tests cost ratios of 5 and 20.

From costs to a decision rule

For a firm with bankruptcy probability $p = P(Y = 1 \mid x)$:

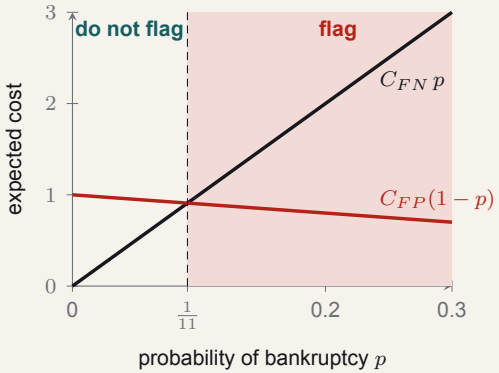
expected cost if we **flag**: $C_{FP}\,(1-p)$

expected cost if we **do not**: $C_{FN}\,p$

Flag when flagging is cheaper:

$$p \;\geq\; \frac{C_{FP}}{C_{FP} + C_{FN}} \;=\; \frac{1}{11} \;\approx\; 0.09$$

Bayes decision rule (Lecture 1).



How we score a model

The **average cost per firm**: the empirical risk under the loss table.

$$\hat{R}_n = \frac{1}{n}\sum_{i=1}^{n} L(\hat{a}_i, y_i) = \frac{10 \cdot \mathrm{FN} + 1 \cdot \mathrm{FP}}{n}$$

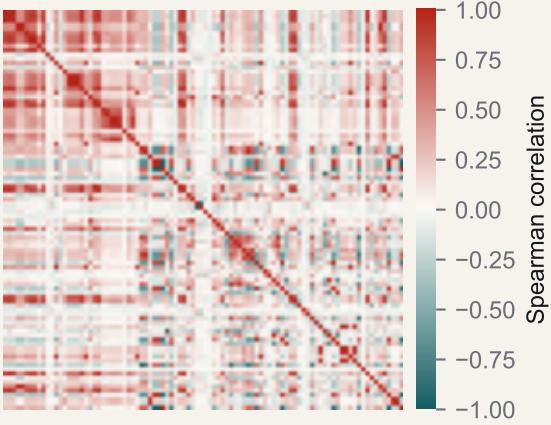
Lower is better. Recall and precision are reported alongside it.

0.32

cost of “never flag” $(10 \times 3.2\,\%)$

— The bar every model must clear.

Many ratios say the same thing



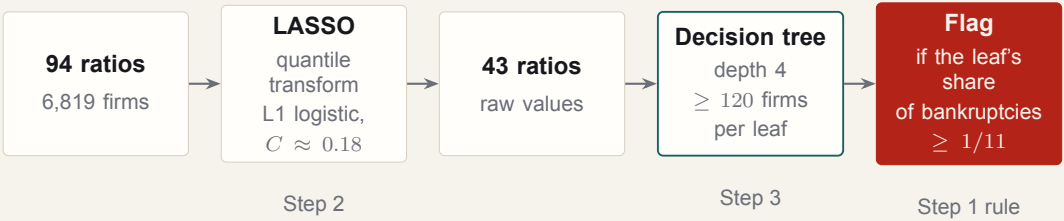
50 pairs

of ratios with $|\rho| > 0.95$, involving 47 of the 94 ratios

Spearman correlation works on ranks, so it also captures a ratio and its inverse.

— A strong case for variable selection.

Our approach in one picture



— Everything is fitted on the training set. The test set is opened once.

PART 02

Variable selection

Which of the 94 ratios do we really need?

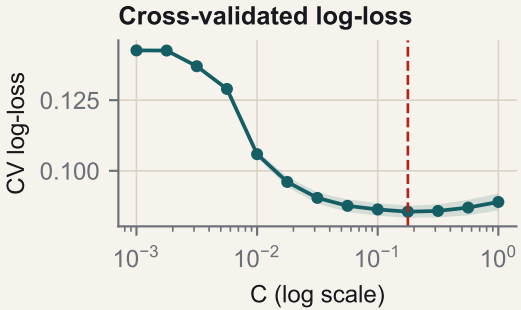
02

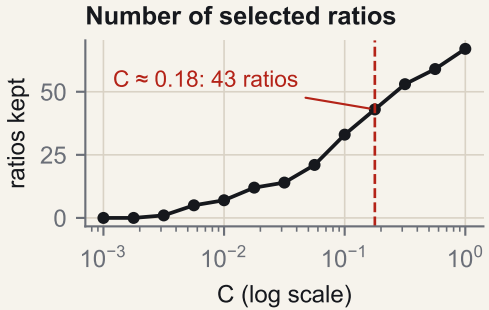
LASSO for a yes/no outcome

$$\min_{b_0,\beta} \underbrace{-\frac{1}{n}\sum_{i=1}^{n}\Big[y_i \log p_i + (1-y_i)\log(1-p_i)\Big]}_{\text{fit to the data}} + \underbrace{\lambda \|\beta\|_1}_{\text{L1 penalty}}$$

- The L1 penalty sets many coefficients to **exactly zero**: those ratios are dropped (Lecture 4).
- **One knob**, λ (or $C = 1/n\lambda$), sets how many ratios survive.
- Fast enough to be **refit inside every cross-validation fold**.

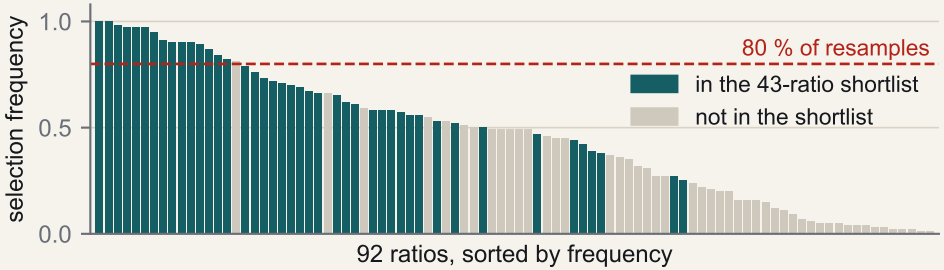
Choosing the penalty





— **Lowest cross-validated log-loss at $C \approx 0.18$: 43 of the 94 ratios are kept.**

Is the shortlist stable?



100 bootstrap refits of the whole procedure. Only 15 of the 43 ratios survive at least 80 % of the time.

Correlated ratios swap places: a predictive shortlist, not a list of causes.

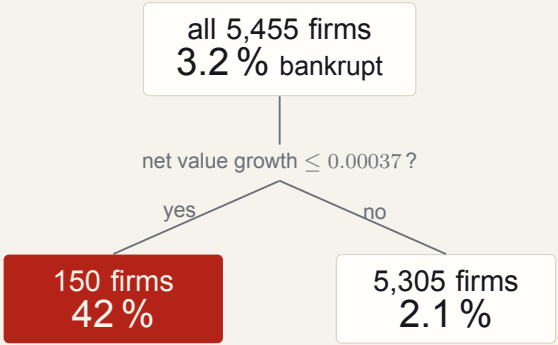
PART 03

Model selection & tuning

How deep should the tree grow, and does selection really help?

03

How a tree flags a firm



Splits make the leaves **purer**, measured by the Gini impurity:

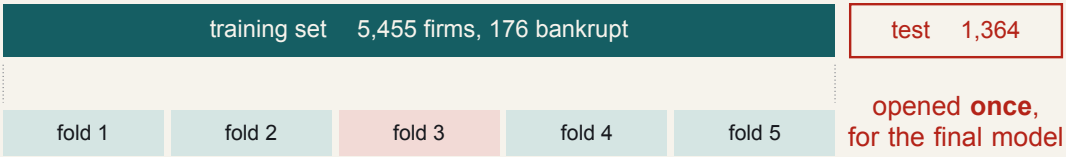
$$G(q) = 2q(1-q)$$

Each leaf's **share of bankruptcies** $\hat{\eta}(x)$ estimates $P(Y = 1 \mid x)$.

— **Flag the firm if** $\hat{\eta}(x) \geq 1/11$**.**

An honest protocol

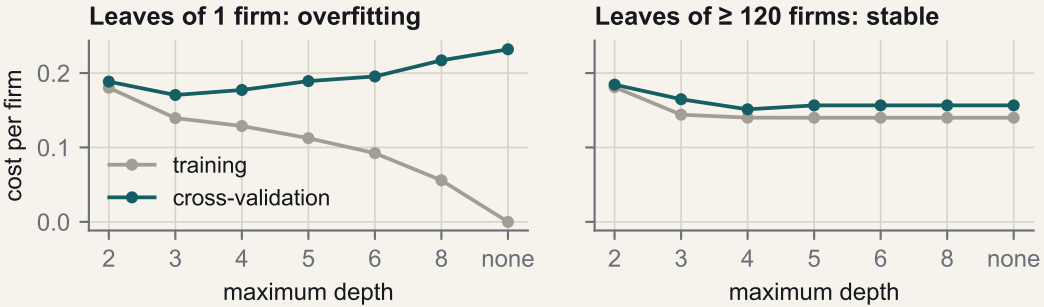
6,819 FIRMS, STRATIFIED 80 / 20 SPLIT



5-fold cross-validation: quantile transform, LASSO and tree are **refit on 4 folds**, scored on the 5th.

— **No decision ever sees the test set.**

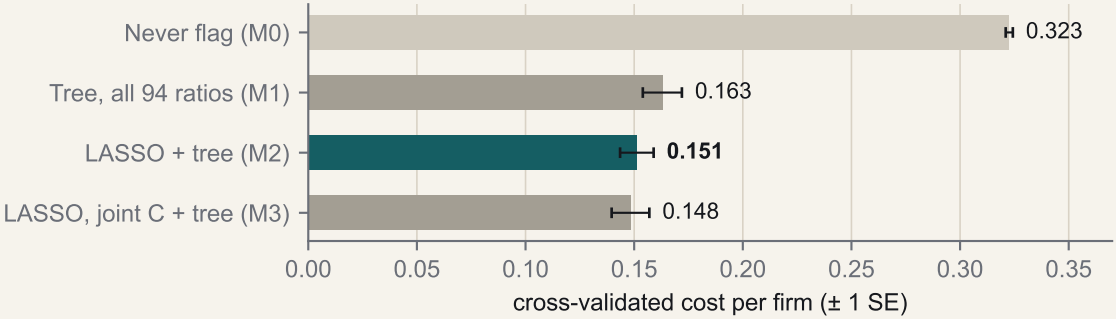
Leaves must be large



Leaves of one firm memorise the data: training cost falls to 0, CV cost climbs to 0.23.

— With at least 120 firms per leaf, depth 4 is best: 0.151 per firm.

Four candidates, same five folds



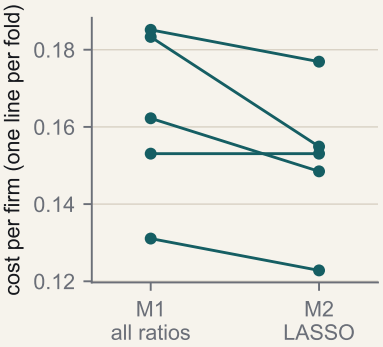
Every tree halves the cost of never flagging.

Does selection really help?

Same folds, so we compare **fold by fold**:

$$\bar{\Delta} \pm \frac{s_\Delta}{\sqrt{5}}, \qquad \Delta_k = \text{cost}_k(\text{M2}) - \text{cost}_k(\text{other})$$

vs M2	gain	verdict
M1, all 94 ratios	-0.012 ± 0.005	never beats M2
M3, joint C (7 ratios)	$+0.003 \pm 0.007$	within noise



— **Selection helps. We keep M2.**

The price of interpretability

OUR CHOICE

One decision tree (M2)

0.151

cross-validated cost per firm

9 leaves, 8 ratios: **every decision is a readable rule.**

BENCHMARK ONLY

A random forest (M4)

0.136

cross-validated cost per firm

300 trees vote: finer probabilities, **but no rule to read.**

— **The forest saves about 10 % (0.015 ± 0.007): the price of readable rules.**

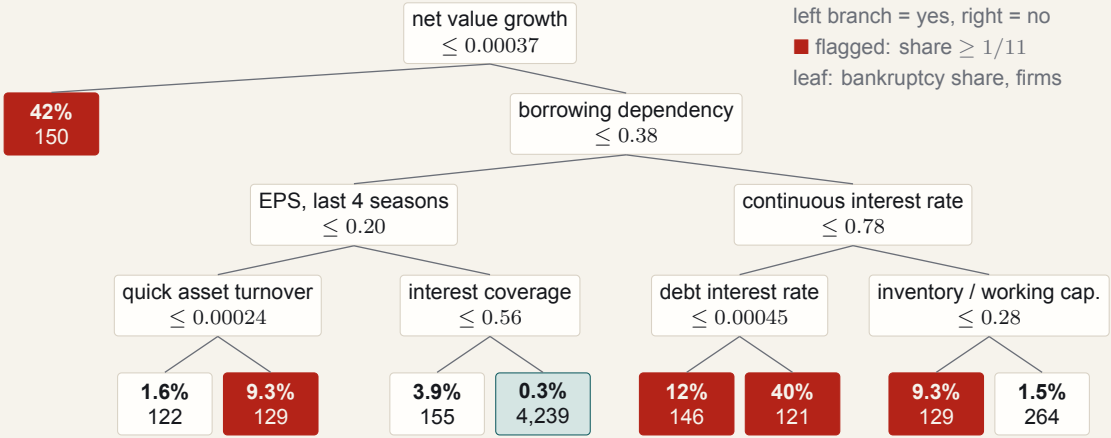
PART 04

Evaluation

One look at 1,364 firms the model has never seen.

04

The final tree



Reading the tree

THREE ALARMS

1. **Weakest net value growth** (bottom 2.7 %): 42 % go bankrupt.
2. **Heavy borrowing + weak profitability**: 40 % when the debt is expensive, 12 % otherwise.
3. **Two narrower pockets** at 9.3 %: flagged, but only just.

ONE SAFE ZONE

Moderate borrowing, EPS above 0.20, interest coverage above 0.56:

0.3 %

bankrupt, in a leaf holding 78 % of the firms

Predictive associations, not causal effects.

Results on unseen firms

0.179 vs 0.323

cost per firm, model vs never flag: **−45 %**

32 / 44

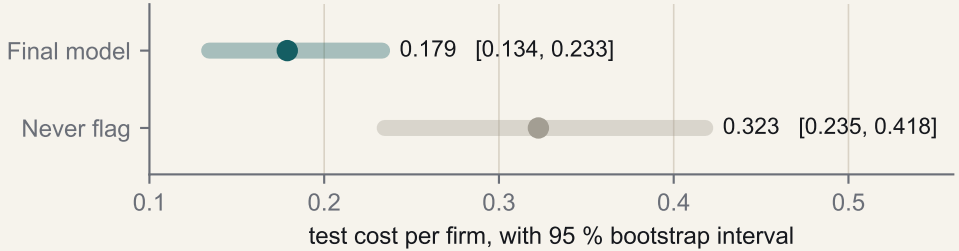
bankruptcies caught

156

firms flagged (1 in 9)

	not flagged	flagged
healthy	1,196 correct	124 false alarms ($\times 1$)
bankrupt	12 missed ($\times 10$)	32 caught

Is the gain real?



Hoeffding (Lecture 2): too wide to decide

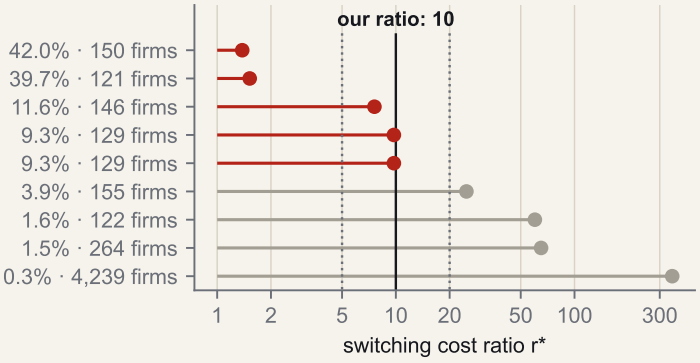
$\pm C_{FN}\sqrt{\log(2/\alpha)\,/\,2m} \;=\; \pm 0.37$

Bootstrap, 1,000 resamples of the test firms:

gain in $[0.065,\ 0.228]$ per firm

— The interval excludes 0: the model really beats not screening.

What if the costs were different?



A leaf is flagged once the cost ratio exceeds

$$r^* = \frac{1 - \hat{\eta}}{\hat{\eta}}$$

Any ratio between 9.75 and 24.8 gives the same decisions.

— **Two leaves sit right at the edge.**

RECOMMENDATION

Our recommendation

1 Action

Flag every firm whose leaf has a bankruptcy share of at least 1/11: 8 ratios, a depth-4 tree.

2 Assumptions

A missed bankruptcy costs 10 false alarms; leaf shares are taken as probabilities.

3 Versus no screening

0.179 vs 0.323 per firm: 32 of 44 bankruptcies caught for 124 false alarms.

4 What would change it

A cost ratio below 9.75 unflags two leaves; below 7.6, a third.

LIMITATIONS

What to keep in mind

- **Assumed costs.** The 10 : 1 ratio is stipulated.
- **No time dimension.** A random split, not a forecast of future years.
- **Repeated firms?** The same firm may appear in several years.
- **Few bankruptcies.** 176 in training, 44 in the test set.
- **Fragile leaves.** Two leaves sit 0.002 above the threshold, within noise (± 0.026).
- **Unstable shortlist.** A 7-ratio selection works about as well.

THANK YOU

Questions?

THREE TAKEAWAYS

— **Costs, not accuracy**: flag when $P \geq 1/11$.

— **LASSO + a small tree**: 45 % lower cost on unseen firms, with readable rules.

— **A forest** is about 10 % cheaper: the price of interpretability.

github.com/yassineerraji/Augmented-Decision-Final-Project

YASSINE ERRAJI · WILLIAM PELTIER · BEN DAESH · SIMON AMAR